Notes for Prof. Chong Liu

Yusuke Ozaki
University at Albany, SUNY, Department of Computer Science
Updated: November 10, 2026

Thank you for agreeing to write for me. This memo collects facts—periods, what I worked on, and episodes with dates—to help you recall our work. It is not a draft letter; what to evaluate and how is entirely yours. Details: ozatamago.github.io/research; CV: ozatamago.github.io/assets/cv.pdf.

Our working relationship

- Research advisor since Aug 2025, when my exchange year at UAlbany began (Aug 2025 – May 2026). Since I returned to Japan (Jun 2026), we have continued online: preparing the LiP-UCB submission together, and weekly meetings on the quantum project.
- Student in your course ICSI 436 Machine Learning (Spring 2026; grade A).
- How we interacted: weekly research meetings, joined by Prof. Chaowen Guan, and discussion on Slack. All meetings and writing were in English.

Research with you

Lipschitz bandits (Feb 2026 – present) — [R2], under review

- **Question.** How well does a classical algorithm (Piyavskii–Shubert) for maximizing an unknown function with few trials perform over all its trials, and what changes when observations are noisy?
- **Without noise.** Proved that the algorithm is as good as any algorithm can be, up to constants, in every dimension.
- **With noise.** Proposed a new algorithm, LiP-UCB, with a guarantee close to the best possible.
- **An existing proof.** Found a problem in a published proof of the best-possible limit and re-derived it.
- **Experiments.** Simulations that agree with the theory.
- **Roles.** You proposed studying the Piyavskii–Shubert algorithm. I derived the results, ran the experiments, and wrote the manuscript, with extensive advice from you on the writing.

Preference-based Bayesian optimization (Aug 2025 – Jan 2026)

- Bayesian optimization that learns from comparisons (which of two options is preferred) instead of numerical scores; I studied what can be guaranteed in theory. It did not work out, and we paused it.

Quantum algorithms for Bayesian optimization (Jun 2026 – present)

- With Chaowen Guan (University of Cincinnati): whether quantum algorithms can speed up the main computation in Bayesian optimization.

Episodes you may remember

1. **Feb 2026 — starting the Lipschitz-bandit project.** You suggested the Piyavskii–Shubert algorithm. We started because it is intuitive and geometrically beautiful, yet well-known papers had not used it.
2. **2026 — writing the paper in English.** I wrote the manuscript and revised it with your extensive advice on the writing. Since Jun 2026 we have prepared the submission together online.
3. **Mar – May 2026 — ICSI 436 course project.** With a classmate (Jason Mauro), extended an existing Bayesian-optimization method for networks of connected functions to active learning, where only some parts of the network are measured. I implemented the methods and ran the experiments.

My undergraduate thesis at Kwansei Gakuin (due Feb 2027) is on this Lipschitz-bandit work.

How the three letters fit together

Schools read the three letters side by side, so it helps if each covers what that recommender saw most closely. My suggestion, which you are free to ignore:

- **Prof. Takahashi:** the main research (OFML, ProtocolUCB) and the lab automation—taking a research question to a method, theory, implementation, and experiments on physical instruments.
- **Prof. Inokuchi:** computer-science foundations and growth through undergraduate research (sparse graph neural networks, the undergraduate thesis).

- **Prof. Liu:** mathematical research (problem formulation, proofs, re-examining lower bounds) and research carried out in English.

Please describe only what you observed directly. My work with IBM (SPIN) and at the University of Osaka is in the CV as background; I am not asking anyone to evaluate it.

If you adjust by program (optional)

Your letter is the closest view of my mathematical research and of research done in English. Most of the programs are in bioengineering, chemical engineering, or the life sciences (BU CompE, Columbia BME, UCSB BioE, Berkeley CBE, UC Riverside CEE, NYU BME, Northwestern IBiS, Cornell Food Science); for these, the statistical modeling and guarantees behind choosing the next experiment (Bayesian optimization, bandits). For Stanford CS: proving guarantees and re-examining existing proofs. The facts stay the same either way.